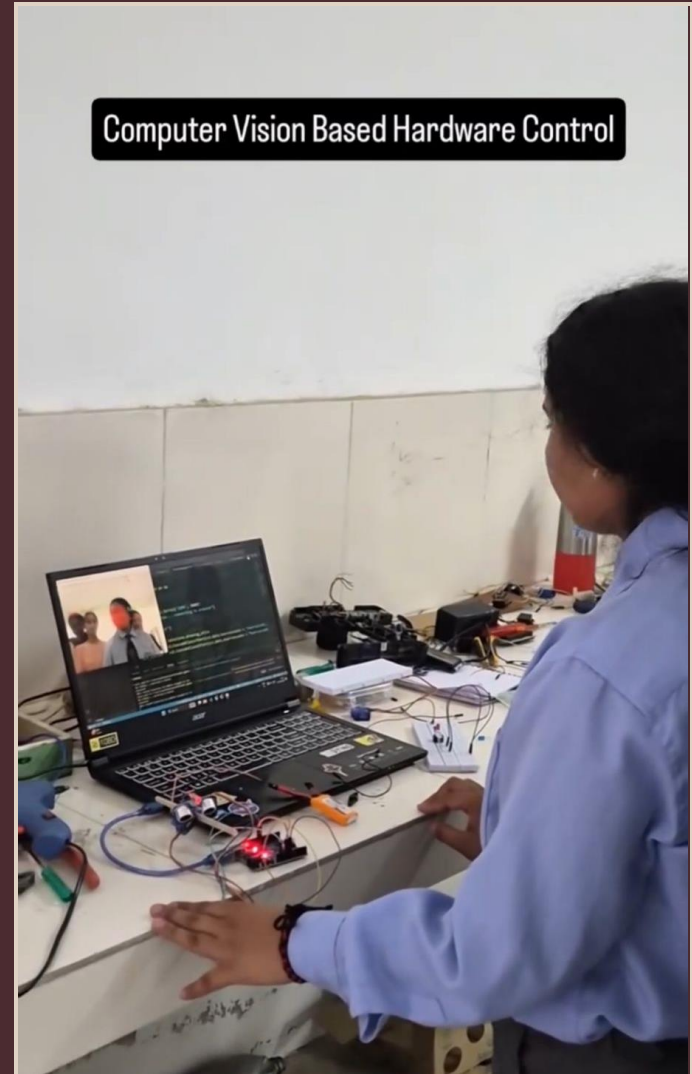


Computer Vision based hardware control

First Up Consultants



Computer Vision Based Hardware Control

is a system where a camera detects an object, gesture, color, or movement and uses that information to control physical hardware such as motors, LEDs, servos, or robots.

Camera » Image Processing » Detection » Decision » Hardware Control



Camera captures the video.



OpenCV/Python processes the image.



The system detects something—for example, a hand gesture, color, or object.



Based on the detection, a command is generated.



Arduino/ESP32 receives the command.



Hardware responds by controlling motors, servos, LEDs, etc.



Main Technologies

1. OpenCV

Used for:

Object detection

Color detection

Face detection

Hand/finger detection

Motion detection

Shape detection

Tracking

2. Arduino / ESP32

Acts as the **controller** that receives commands from the computer and operates the hardware.

3. Motors & Servos

Used to create physical movement.

Examples:

DC motor » wheels

Servo motor » robotic arm

Stepper motor » precise movement

4. Communication

The computer can send commands to the microcontroller through:

USB/Serial

Bluetooth

Wi-Fi